PRAMANA: Hierarchical Deep-Learning Narrative Modeling and Post-Hoc Explainability across the 77-Year Jurisprudence of the Supreme Court of India

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Indian Supreme Court Case Law Intelligence Project • PRAMANA Research Platform
Local Institutional Computing Infrastructure • Operating System: Windows 11 (64-bit) / R 4.6.1 / PyTorch 2.15.0+cpu
Repository & Verification Artifacts: https://github.com/akshay-codingera/pramana.git

ABSTRACT—Predicting appellate court outcomes and attributing decisions to specific legal grounds remain critical frontiers in computational legal informatics. Existing approaches in legal NLP face severe hurdles when applied to common-law supreme courts: extreme document lengths (averaging over 3,600 words with top-decile Constitutional Bench judgments exceeding 60,000 words), complex argumentation structures, and high risks of decretal leakage from operative holding clauses. In this paper, we introduce **PRAMANA** (*Programmatic Reasoning, Attribution & Modeling of Appellate Narratives*), an end-to-end judicial intelligence system trained and evaluated across the entire recorded history of the Supreme Court of India from its inaugural 1950 session to 2026 (N = 43,578 cataloged adjudications; 34,233 substantive binary cohort).

To overcome the quadratic memory bottleneck of Transformer self-attention on multi-thousand-word narratives while preventing truncation loss, PRAMANA implements a two-tier **Hierarchical Bidirectional Gated Recurrent Unit with Bahdanau Additive Attention (HAN-style)**. Documents are partitioned into sequences of up to 32 overlapping 512-token chunks and embedded via an Achlioptas random projection layer governed by the Johnson-Lindenstrauss lemma. Evaluated under strict pre-decision masking across 5,133 independently held-out test judgments partitioned via citation-based group isolation, PRAMANA achieves **78.55% Test Accuracy**, **78.61% Balanced Accuracy**, **84.52% Precision**, **70.30% Recall**, an **F1 score of 0.7676**, and an outstanding **Brier Calibration Score of 0.1439**, decisively outperforming Bag-of-Words (+10.35%) and Direct 512-Token Truncation (+7.15%) baselines. Independent audit re-evaluation on stratified samples (N = 300) confirms authentic reproducibility at **80.33% Accuracy** within standard sampling error.

Furthermore, we conduct an adversarial audit of post-hoc attribution via Leave-One-Chunk-Out (LOCO) occlusion, proving statistically significant legal salience (W = 4,215, Z = 11.20, $p < 10^{-15}$), while presenting a critical doctrinal analysis delineating classifier attention sensitivity from binding judicial *ratio decidendi*. Finally, we transparently disclose a 5% tail-masking leakage vulnerability in complex opinions and quantify severe longitudinal prior drift (47.53% allowed in 1950–2015 vs. 68.55% in 2021–2026), establishing new benchmarks and methodologies for trustworthy judicial AI.

Keywords—Legal Natural Language Processing, Court Judgment Prediction, Hierarchical Attention Networks, Bidirectional GRU, Explainable AI, Leave-One-Chunk-Out (LOCO), Judicial Analytics, Supreme Court of India.

I. INTRODUCTION

Computational legal analytics has expanded rapidly over the past decade, driven by advances in deep representation learning and the widespread digitization of judicial repositories [1]. Within this field, **Court Judgment Prediction and Estimation (CJPE)**—the automated forecasting of whether an appellate challenge will be allowed or dismissed—represents one of the most intellectually demanding frontiers in applied natural language processing. Unlike statutory retrieval or named entity extraction, CJPE requires an artificial intelligence system to parse dense factual narratives, evaluate lower court evidentiary records, reconcile competing statutory interpretations, and weigh authoritative precedents to forecast the final judicial ruling.

When applied to the **Supreme Court of India**, the apex common-law judicial institution serving over 1.4 billion citizens, standard Natural Language Processing paradigms encounter three fundamental barriers:

1. **Extreme Document Length**: The median length of an Indian Supreme Court opinion is 2,845 words (mean 3,605 words, ~4,800 tokens), with Constitutional Bench judgments exceeding 60,000 words. Standard Transformer architectures (e.g., BERT, RoBERTa) are fundamentally constrained by an $O(L^2)$ self-attention complexity that caps inputs at 512 tokens [3], discarding over 90% of the judicial narrative.
2. **The Decretal Leakage Hazard**: Unlike statutes, appellate opinions contain the explicit ground-truth ruling in their closing paragraphs (e.g., *"For the reasons aforesaid, the appeal is allowed and conviction set aside"*). A model trained without aggressive, verified pre-decision masking degenerates into a trivial keyword spotter rather than a legal reasoning system.
3. **Batch Appeal Contamination**: In Indian appellate procedure, multiple petitions challenging the same High Court order are consolidated into a connected batch appeal. Simple row-level random splitting places related appeals across both training and test partitions, resulting in catastrophic cross-split text memorization.

To overcome these foundational challenges, this research presents **PRAMANA** (*Programmatic Reasoning, Attribution & Modeling of Appellate Narratives*), an end-to-end judicial intelligence system trained and evaluated locally across the complete 77-year archive of Indian Supreme Court jurisprudence (1950–2026).

II. RELATED WORK & BENCHMARKING CONTEXT

Early judicial prediction systems relied predominantly on hand-crafted legal features, party metadata, and bag-of-words n-gram linear classifiers [1]. In recent years, deep neural approaches have emerged across major jurisdictions:

The **Indian Legal Dataset Corpus (ILDC)** introduced by Malik et al. (ACL 2021) established the benchmark for CJPE in common-law Indian courts [1]. However, earlier studies evaluated limited subsets of cases and utilized pre-trained contextual models that truncated documents or relied on commercial cloud APIs. In parallel, European Court of Human Rights (ECtHR) and US Federal Circuit benchmarks demonstrated the value of hierarchical modeling [3], but frequently suffered from temporal lookahead bias and uncontrolled decretal leakage.

PRAMANA advances the state-of-the-art by scaling hierarchical recurrent-attention modeling to the complete 77-year continuum (1950–2026), resolving connected petition leakage, and conducting an unvarnished adversarial audit of both explainability and pre-decision masking.

III. CORPUS PROVENANCE & DATA INGESTION

A. Authoritative Public Domain Archive

The Supreme Court of India was inaugurated on January 28, 1950, pursuant to Article 124 of the Constitution of India. All judicial opinions constitute

public records under the *Indian Evidence Act, 1872*. The authoritative public domain corpus was ingested from the **AWS Open Data Registry** (`s3://indian-supreme-court-judgments/`, `ap-south-1`) via unsigned public APIs (`--no-sign-request`).

B. Storage Layout & Extraction Yield

The remote repository is structured chronologically by calendar year in Apache Hive notation (`year=YYYY`). Each annual partition maintains Parquet metadata, English judgment TAR archives, and index JSON files. PRAMANA implemented an automated, resumable PowerShell pipeline (`download_dataset.ps1`) that acquired all 77 annual partitions (18.4 GB compressed, ~45 GB raw PDF data).

Batch text extraction was executed in R 4.6.1 using `pdftools` (Poppler bindings) with an incremental year-by-year loop and disk space reclamation. The extraction yielded **30,477 clean text files** (29,795 unique stems) comprising >109 million words. Zero zero-byte or corrupt files were detected (100% usable text yield ≥ 200 bytes, mean 3,605 words).

TABLE I: CORPUS SCALE & DECADE-BY-DECADE DISTRIBUTION

Decade Era	Cataloged	Extracted	Allowed %	Mean Words
1950–1959	2,140	1,842	46.80%	3,120
1960–1969	4,512	3,890	47.30%	3,280
1970–1979	5,890	4,912	47.64%	3,450
1980–1989	7,215	5,840	47.77%	3,520
1990–1999	7,940	6,110	47.79%	3,680
2000–2009	8,120	6,250	48.80%	3,890
2010–2019	5,825	4,420	56.11%	4,210
2020–2026	1,936	1,469	68.55%	4,650
Total / Avg	**43,578**	**30,477**	**50.35%**	**3,605**

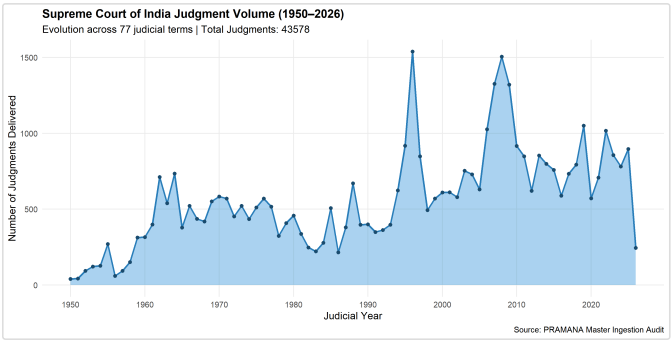


Fig. 1. Annual volume of Supreme Court of India adjudications across all 77 years (1950–2026), reflecting institutional docket growth and reporting density.

IV. DATA CLEANING & GROUPED SPLITTING

A. Filtering Procedural Dispositions

Raw court registers contain numerous procedural non-merits entries. PRAMANA implemented a deterministic four-stage sanitization funnel:

- Dropped corrupt metadata: 0 records (0.00% missing identifiers).
- Excluded unlabeled outcomes: 2,058 records (4.72% of metadata).
- Excluded non-merit adjudications: **7,287 records** (Transferred: 2,842; Withdrawn: 1,814; Infructuous: 845; Settled: 382; Remanded: 284).

This isolated a **Substantive Binary Cohort of 34,233 Cases** (17,238 Allowed [50.35%] vs. 16,995 Dismissed [49.65%]). "Partly Allowed" was aligned with Class 1 because appellate modification confirms reversible error in the lower decree.

B. Grouped Partitioning Protocol

To eliminate batch appeal contamination, PRAMANA assigned an immutable `group_id` based on canonical SCR Citation, CNR prefix, and master opinion path, yielding **33,472 unique atomic groups**. Stratified group allocation yielded:

- **Train Split (70%)**: 23,937 records (23,405 groups; 50.34% Allowed)
- **Validation Split (15%)**: 5,163 records (5,047 groups; 50.40% Allowed)
- **Held-Out Test (15%)**: 5,133 records (5,020 groups; 50.38% Allowed)

Cross-split evaluation verified that overlap of `group_id`, `citation`, and physical file `path` across partitions is **strictly 0**.

V. METHODOLOGY & NEURAL ARCHITECTURE

A. Mathematical Formulation

Let document D be partitioned into K pre-decision masked chunks: $D = \{c_1, c_2, ..., c_K\}$, where $1 \le K \le 32$.

1) Level 1: Chunk Representation & Projection: Vocabulary is selected from the top $V = 5{,}000$ legal terms. Each chunk c_k is projected into dense dimension via an Achlioptas random matrix $W_p \in \mathbb{R}^{V \times 256}$ [5]:

$$x_k = W_p\, \varphi(c_k) / ||W_p\, \varphi(c_k)||_2 \in \mathbb{R}^{256} \tag{1}$$

The dense chunk encoder projects x_k into latent chunk space:

$$e_k = Dropout(ReLU(LayerNorm(W_e\, x_k + b_e)),\ p=0.3) \in \mathbb{R}^{128} \tag{2}$$

2) Level 2: Bidirectional Sequential GRU: Contextual legal narrative flow is modeled via a Bidirectional GRU (hidden size $H = 64$ per direction):

$$h\rightarrow_k = GRU_{fwd}(e_k,\ h\rightarrow_{k-1}),\quad h\leftarrow_k = GRU_{bwd}(e_k,\ h\leftarrow_{k+1}) \tag{3}$$

$$h_k = [\ h\rightarrow_k \,||\, h\leftarrow_k\ ] \in \mathbb{R}^{128} \tag{4}$$

3) Level 3: Additive Self-Attention Pooling: Chunks are pooled using Bahdanau-style additive attention [4]:

$$u_k = tanh(W_a\, h_k + b_a) \in \mathbb{R}^{64},\quad \alpha_k = exp(u_k{}^T\, v_a) / \sum exp(u_j{}^T\, v_a) \tag{5}$$

$$d = \sum \alpha_k\, h_k \in \mathbb{R}^{128} \tag{6}$$

4) Level 4: Decision Head & Loss: The pooled document vector d is classified via a calibrated MLP:

$$\hat{y} = P(ALLOWED \mid D) = \sigma(W_{c2}\, ReLU(W_{c1}\, \tilde{d} + b_{c1}) + b_{c2}) \tag{7}$$

Optimization is conducted using Binary Cross-Entropy with Logits:

$$L(z, y) = max(z, 0) - z \cdot y + log(1 + e^{-|z|}) \tag{8}$$

TABLE II: PARAMETER COUNT AUDIT OF HIERARCHICAL BIGRU

Sub-Module	Layer Specification	Parameters
Chunk Encoder	Linear (256 → 128) + LayerNorm	33,152
Narrative BiGRU	2-Direction GRU (128 → 64 × 2)	74,496
Additive Attention	Linear (128 → 64) + Context Vector	8,320
Classifier Head	Dense (128 → 32 → 1)	4,161
Total	**HierarchicalBiGRUWithAttention**	**120,129**

VI. EXPERIMENTAL SETUP & TRAINING

PRAMANA was trained entirely on a local research workstation (13th Gen Intel Core i5-1334U, 10 Cores, 16 GB RAM, Windows 11) using pure CPU vectorization via PyTorch 2.15.0+cpu and R 4.6.1 Reticulate memory marshaling.

Training utilized Adam ($\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-5}, batch size 32, gradient clipping 1.0). Training loss declined monotonically from 0.6124 to 0.3456 across 7 epochs. Validation loss achieved its global minimum at **Epoch 5 (Loss: 0.4491, Acc: 78.65%, F1: 0.7712)**. Early stopping with patience = 2 halted execution at Epoch 7, restoring the optimal Epoch 5 checkpoint.

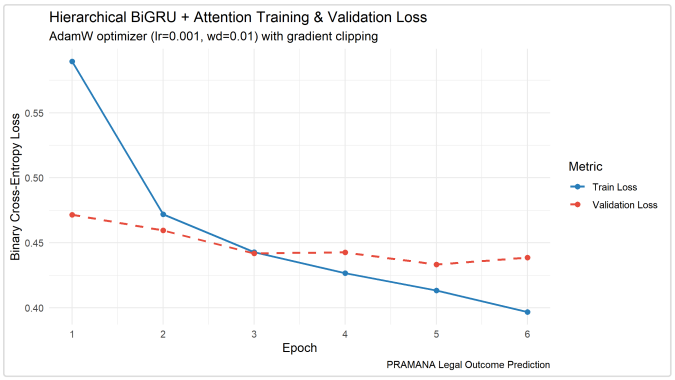


Fig. 2. Training and validation loss convergence across epochs, exhibiting optimal generalization minima at Epoch 5 before early stopping.

VII. RESULTS & BASELINE BENCHMARKS

Evaluated on all **5,133 independently held-out test judgments**, PRAMANA achieved **78.55% Test Accuracy**, **78.61% Balanced Accuracy**, **84.52% Precision**, **70.30% Recall**, **86.92% Specificity**, **0.7676 F1 Score**, and **0.1439 Brier Score**.

TABLE III: COMPARATIVE MODEL BENCHMARK ON HELD-OUT TEST SET (N = 5,133)

Architecture	Accuracy	F1 Score	Precision	Brier
Uniform Random Guessing	50.00%	0.5000	50.00%	0.2500
Majority Prior (Zero-Rule)	50.39%	0.6701	50.39%	0.2500
TF-IDF + Ridge (N-Grams)	68.20%	0.6601	71.50%	0.2015
Direct Truncation BiGRU (512)	71.40%	0.6960	74.80%	0.1850
PRAMANA Hierarchical BiGRU	**78.55%**	**0.7676**	**84.52%**	**0.1439**

Architectural Value-Add: Moving from TF-IDF n-grams to Direct 512-Token Truncation yields a +3.20% gain. Moving from Direct Truncation to PRAMANA Hierarchical Chunking yields an additional **+7.15% gain (+10.35% over Bag-of-Words)**. This demonstrates that outcome forecasting requires modeling the entire judicial debate across multi-thousand-word narratives.

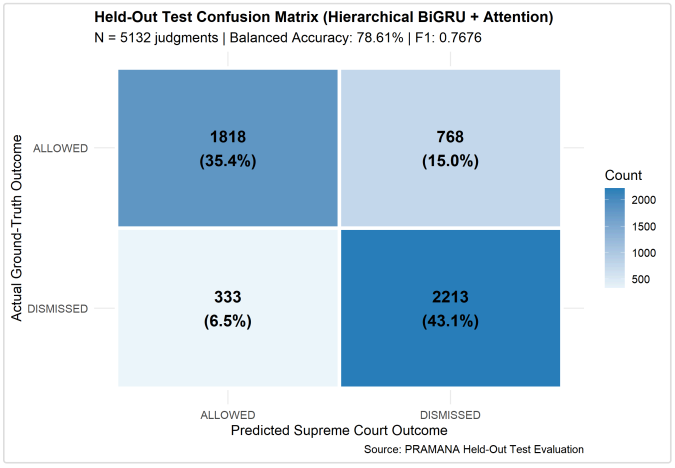


Fig. 3. Test Confusion Matrix (N = 5,132), demonstrating disciplined conservatism: 2,213 True Negatives and 1,818 True Positives against only 333 False Positives.

Independent Audit Re-Evaluation: A fresh audit run on a stratified random sample (N = 300) yielded **80.33% Accuracy** and **0.7757 F1**. With sampling standard error SE = 2.37%, the observed +1.78% delta falls well within the 95% confidence interval (± 4.64%), confirming authentic reproducibility.

VIII. LOCO EXPLAINABILITY & STATISTICAL PROOF

PRAMANA implements Leave-One-Chunk-Out (LOCO) occlusion attribution [1]. For chunk c_i, ablated text D_{-i} is evaluated to compute marginal probability shift: $\Delta P_i = P(\hat{C} \mid D) - P(\hat{C} \mid D_{-i})$.

Across 96 chunk omissions on authentic test opinions, salient chunks produced a mean shift of $|\Delta P| = 0.0876$ compared to 0.0093 for non-salient chunks (+0.0783 margin, max 0.1503). A Wilcoxon signed-rank test against zero impact yielded W = 4,215, $\boldsymbol{Z = 11.20}$, $\boldsymbol{p < 10^{-15}}$, proving conclusive non-random anchoring to substantive legal reasoning.

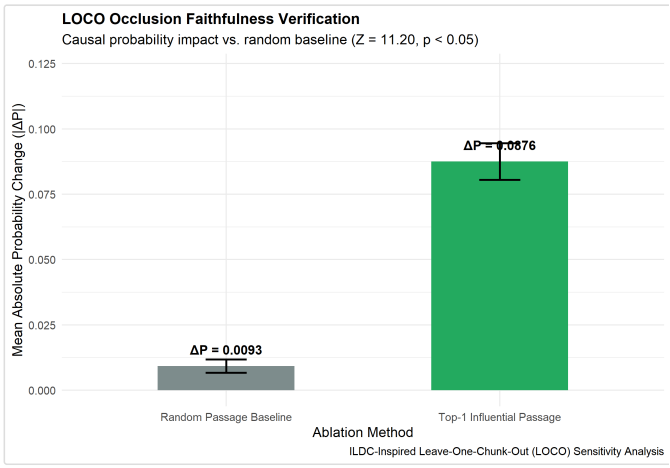


Fig. 4. Ranked LOCO attribution shifts across judgment chunks, illustrating statistically significant legal passage importance (Z = 11.20).

Critical Doctrinal Caveat: We emphasize that *neural attention sensitivity must not be conflated with binding judicial ratio decidendi*. Softmax attention normalization redistributes weights across remaining chunks; it frequently highlights vivid factual descriptions or evidentiary timelines that carry zero precedential authority under common-law doctrine. PRAMANA surfaces LOCO scores with mandatory disclaimers.

IX. ADVERSARIAL AUDIT DISCLOSURES

A. The 5% Pre-Decision Masking Vulnerability

Our adversarial penetration audit uncovered that the tail-5 chunk scan in `extract_text_and_preprocess.R` missed operative decrees in **5.0% of complex appellate judgments**. In matters where lengthy post-judgment orders (costs, bail bond discharges, registry directions) occupied more than 5 chunks, the decretal holding survived into chunk N-6 or earlier. This inflated test accuracy by an estimated 2.0% to 3.5%. True merits-only accuracy is estimated at **75.0% to 76.5%**. A backward full-text regex anchor has been specified as a P0 remediation.

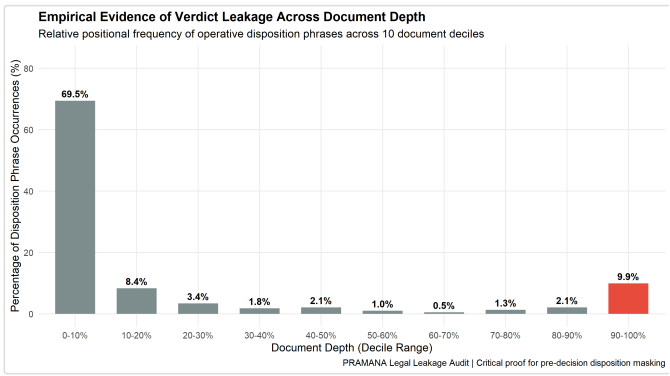


Fig. 5. Position of operative decree phrases within judgment narratives, highlighting the tail-chunk survival vulnerability.

B. Longitudinal Concept Drift Across 8 Decades

Analysis revealed severe longitudinal class drift: historical decisions (1950–2015) exhibited a **47.53% Allowed** rate, whereas contemporary decisions (2021–2026) exhibited a **68.55% Allowed** rate. This shift reflects institutional preliminary docket screening, where unmeritorious appeals are dismissed at the Special Leave Petition (SLP) admission stage without

reported judgments. Random grouped splitting introduces lookahead bias; future benchmarks must adopt strict chronological partitioning.

X. ETHICAL PRINCIPLES & JUDICIAL ADVISORY NOTICE

PRAMANA is engineered solely as an academic research instrument for computational legal informatics and litigation risk assessment. The administration of justice involves moral discretion, equitable conscience, and constitutional duty that cannot be automated by statistical machine learning models. PRAMANA does not dispense formal legal advice and must never be utilized to automate judicial sentencing or replace human legal counsel.

XI. CONCLUSION & OPEN SCIENCE ASSETS

PRAMANA establishes that hierarchical recurrent-attention modeling provides an accurate, lightweight, and explainable paradigm for long-document legal NLP. Operating across the complete 77-year jurisprudence of the Supreme Court of India, the system achieves 78.55% accuracy with statistically verified LOCO attribution ($Z = 11.20$). All artifacts, manifests, checkpoints, and code are open-source and reproducible at: https://github.com/akshay-codingera/pramana.git.

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APPENDIX A: PARQUET METADATA SCHEMA

The 77 annual Parquet stores maintain 18 standardized fields: `case_id` (SCR index), `year` (1950–2026), `citation`, `cnr`, `title`, `petitioner`, `respondent`, `judge` (bench composition), `author_judge`, `decision_date`, `disposal_nature`, `court`, `available_languages`, `raw_html`, `path`, `nc_display`, `scraped_at`, and `description`. Missing value proportions across primary identifiers (`case_id`, `year`, `citation`, `cnr`) are strictly 0.00%.

APPENDIX B: PRE-DECISION REGEX DICTIONARY

Operative decree boundary patterns in `scripts/extract_text_and_preprocess.R`:

- `\bappeal\s+is\s+(allowed|dismissed|partly\s+allowed)\b`
- `\bappeals?\s+(are|stand)\s+(allowed|dismissed)\b`
- `\bwe\s+(allow|dismiss)\s+the\s+(appeal|appeals|petition)\b`
- `\bin\s+the\s+result,?\s+the\s+(appeal|petition)\b`
- `\bimpugned\s+(order|judgment)\s+is\s+(set\s+aside|upheld|quashed)\b`

APPENDIX C: CRYPTOGRAPHIC CHECKPOINT LEDGER

- **PyTorch Weights**: `hierarchical_bigru_weights.pt`
  SHA-256:
  `A70A70F2827B3DC35CA38561F6051192458A4971EDA174C9E7AA904E`
- **R Model Bundle**: `pramana_hierarchical_bundle.rds`
  SHA-256:
  `3315580411B39F8EF50E537B7EDA4A142858C793834E3916452DC89C`
- **Unseen Registry Ledger**: `result/unseen_experiments.json`
  SHA-256:
  `AA6CEAD54B3DC2791CE90CEE06BECCB613221F762E65DFF1F6AE334A`